

# Sebastian Lopez

sebastian.t.lopez@berkeley.edu | (415) 629-2227 | linkedin.com/in/sebastiantlopez/

## Education

### University of California, Berkeley

Expected: May 2028

Bachelor of Science in Mechanical Engineering

- **Coursework:** Mechanical Behavior of Engineering Materials, Mechatronics Design, Fluid Mechanics, Thermodynamics, Three-Dimensional Modeling, Manufacturing and Design Communication, Solid Mechanics, Multivariable Calculus
- **Cal Theta Tau** (Co-ed Professional Engineering Fraternity): Coordinating fundraising initiatives including restaurant partnerships & football games concessions for chapter activities, orchestrating sponsorships with companies (Polymarket)

## Professional Experience

### Mechanical Design Engineer

Berkeley, CA

ClearPath Medical

Sep 2025 – Present

- Initiated development of scalable in-house connector platform to reduce supplier reliance and procurement lead times.
- Designed high-voltage connector platform with IEC 60601 and MOOP compliance for medical protection standards.
- Performed conceptual design and detailed analysis addressing high-voltage creepage and clearance safety challenges.
- Researched advanced composite material solutions to improve overall electrical insulation and prevent conductor arcing.
- Evaluated scalability in pin count and optimized for compact low-voltage and transient high-voltage surge applications.

### Civil Engineering Intern

San Francisco, CA

PGH Wong Engineering

Jun 2025 – Oct 2025

- Contributed to rail alignment optimization following scope reduction from Mountain House terminus, analyzing alternative pathways and Z-curve configuration at Vasco intersection to efficiently connect with existing ACE railway station.
- Developed parcel inventory from Alameda county assessor data including ownership details, public/private classifications, and footprint calculations by easement type to support RoW acquisition and preliminary engineering documentation.
- Assisted in developing highway connection layouts for the Livermore–Vasco corridor, mapping Right-of-Way exhibit strategies and defining aerial easement pathing for the freeway breakaway section initiating the Z-curve to the station.

### HS Explainer

San Francisco, CA

The Exploratorium

May 2023 – Aug 2024

- Performed 10 unique science demonstrations engaging visitors and museum guests across a range of scientific fields.
- Trained around 30 incoming High School Explainers every semester on museum operations and scientific demonstrations.
- Facilitated over 1000 visitor interactions daily across 600 exhibits/coordinated with a team of 30+ explainers per shift.
- Utilized a museum-wide radio system to assist with general operations, such as reuniting lost visitors with their families.
- Participated in daily career development training (200+ hours), advancing skills in public speaking and leadership.

### Civil Engineering & Design Intern

San Mateo, CA

PGH Wong Engineering

May 2023 – Jul 2023

- Partnered with 80+ engineers on \$2.6B SFO Terminal 3 expansion/modernization, increasing terminal capacity by 30%.
- Developed phasing/sequencing procedures during construction and minimizing disruptions in active zones by over 50%.
- Executed installments of temporary walkways and refined passenger routing strategies to maintain operational efficiency.
- Honed skills in construction operations, project management, and budget oversight for a large-scale infrastructure project.

## Projects

### Dual-Chamber Plant Drainage System | Mechanical Designer

Feb 2026 - May 2026

- Conceptualized and designed an efficient, working prototype dual-chamber indoor planter system utilizing a perforated separation plate to isolate soil from a bottom runoff reservoir equipped with a user-controlled mechanical drainage tap.
- Developed 3D CAD assemblies and working technical drawings, implementing standard ANSI fit classifications including LN2/LN3 Snap Fits for the lid, RC7/RC8 Loose Running Fits for the reservoir slide, and FN1 Friction Fits for the pump.
- Configured GD&T specifications with critical linear tolerances of  $\pm 0.01$  inches to ensure precise FDM component mating.
- Conducted a material and process selection study comparing FDM and Injection Molding, ultimately utilizing PETG to rapidly execute prototyping and test prints due to its high chemical resistance and cost-efficiency during iterative testing.
- Fabricated and assembled final physical prototypes, validating mechanical valve sealing integrity and overall fitment.

### Wind Turbine & Structural Support Tower | Mechanical Designer

Feb 2026 - May 2026

- Designed and parametrically modeled a horizontal-axis wind turbine assembly in SolidWorks, optimizing the 3.0-inch blade profiles with a 15-degree attack angle and a 20-degree twist angle to maximize kinetic-to-electrical energy conversion.
- Conducted structural FEA using a 35 MPa yield strength ABS material profile, applying a 9.8 N static load to simulate real-world conditions, which predicted a 0.59 mm maximum displacement and verified a robust structural Factor of Safety.
- Fabricated assembly via FDM 3D printing, utilizing an open truss tower design to minimize weight and production cost.
- Executed empirical performance testing in a 25 mph wind stream, successfully producing a peak output of 870 mW.

### Vending Machine | Hardware & Mechanical Lead

Feb 2025 - May 2025

- Designed precise CAD models and detailed prototypes to support accurate fabrication of key structural components.
- Fabricated and assembled frame using precision laser-cut acrylic sheets, ensuring structural stability and accurate fit.
- Collaborated with the electrical team to integrate servo dispensing system, button controls, and LED lighting features.
- Enhanced manufacturability and overall assembly process through design refinements and optimized material selection.
- Incorporated Arduino, embedded, and motor control systems into the frame, aligning mechanical & electrical components.

## Skills

Python, MATLAB, AutoCAD/Modeling, BlueBeam, FEA, 3D Printing, Vectorworks, CNC Routing, SolidWorks, Excel, Revit